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Expedition Objectives

The Bell Island Battle of the Atlantic Expedition is designed to document, interpret, and make accessible one of the most intact Second World War naval battle landscapes in North America.

The project brings together underwater archaeology, marine science—including marine biology and ecological study—advanced imaging technologies, and historical research to produce a comprehensive digital record of the 1942 attacks and their wider operational and geographic context within the Battle of the Atlantic, while also commemorating those who made the ultimate sacrifice and supporting the responsible development of heritage tourism and public interpretation associated with the Battle of the Atlantic in Newfoundland and Labrador.

1. Creation of a High-Resolution Digital Terrain Model (DTM)

At the core of the expedition is the development of a fully integrated three-dimensional Digital Terrain Model (DTM) of the Bell Island battlefield.

This model will combine:

- High-resolution photogrammetry of all four wrecks:
 - *Saganaga* (British)
 - *Lord Strathcona* (Canadian)
 - *Rose Castle* (Canadian)
 - *PLM-27* (Free French under British wartime control)
- Multibeam and acoustic mapping of the surrounding seafloor
- Precise georeferencing using USBL positioning systems
- Aerial drone imaging conducted by a licensed operator from Memorial University's Marine Institute, providing high-resolution coverage of:
 - The wreck sites and surrounding waters in Conception Bay
 - Scotia Pier and the former loading infrastructure
 - The Seamen's Cemetery at Lance Cove
 - The Cemetery Hill / Coastal Road artillery battery
 - The Bell Island Community Museum and No. 2 Mine
- Terrestrial laser scanning of:
 - The Cemetery Hill coastal artillery battery
 - The remains of Scotia Pier
 - The Seamen's Cemetery at Lance Cove

The result will be a unified digital reconstruction of the battlefield—linking seabed, shoreline, and land-based wartime infrastructure into a single coherent spatial model.

2. Underwater Archaeological Documentation

The expedition will conduct systematic, non-intrusive archaeological recording of the wrecks.

This includes:

- Structural documentation of hulls, cargo remains, and damage patterns
- Identification of torpedo impact points and blast effects
- Assessment of site integrity and long-term preservation conditions
- Comparative analysis between wrecks from the September and November 1942 attacks

All work will be conducted in accordance with established underwater cultural heritage protocols, ensuring that the sites are documented but not disturbed.

The expedition is undertaken in consultation with the Shipwreck Preservation Society of Newfoundland and Labrador, which plays an important role in the stewardship and protection of underwater cultural heritage in the province. This collaboration ensures that all documentation is conducted in accordance with established heritage practices and reflects the interests of the local and diving communities.

3. Marine Ecology and Environmental Baseline Study

After more than eighty years on the seabed, the wrecks have become established artificial reef systems within Conception Bay.

The expedition will document:

- Species presence and biodiversity associated with each wreck
- Structural habitat complexity and ecological variation between sites
- Indicators of environmental change, including water temperature and biological shifts

This work will establish a baseline dataset for long-term environmental monitoring, contributing to a broader understanding of marine ecosystems and environmental change in Newfoundland waters.

4. Technology Demonstration and Method Development

A key objective of the expedition is to evaluate and refine advanced marine imaging technologies in a complex, cold-water environment.

This includes:

- Diver-based photogrammetry using high-resolution imaging systems
- Deployment of stereo camera platforms and autonomous or semi-autonomous survey systems
- Integration of optical, acoustic, and positioning data into a unified geospatial model

Particular emphasis will be placed on comparing diver-acquired datasets with those generated by remotely operated and autonomous systems, informing future best practices in underwater documentation and digital reconstruction.

5. Integration of Sea, Shore, and Industrial Landscape

Unlike most naval engagements, the Bell Island attacks occurred within a defended industrial harbour.

The expedition will therefore integrate:

- The wrecks on the seabed
- Coastal defensive positions
- Industrial infrastructure linked to iron ore loading and export
- The surrounding community landscape

This approach treats Bell Island not simply as a collection of wrecks, but as a complete wartime system in which industry, geography, and naval conflict intersect.

6. Public Interpretation and Digital Access

A central goal of the project is to make this history accessible to Canadians and an international audience.

Outputs will include:

- An interactive 3D digital model accessible online
- Visualizations suitable for museum and educational use
- Interpretive materials linking the Bell Island site to the wider Battle of the Atlantic
- On-site interpretive potential, including QR-linked digital access points

In addition to digital access, the project supports the responsible development of heritage tourism associated with the Battle of the Atlantic in Newfoundland and Labrador. By linking Bell Island to key wartime sites across the province—including St. John’s Harbour and the Crow’s Nest Officers’ Club, Signal Hill, Cape Spear, the naval and air base at Argentia, and the transatlantic aviation hub at Gander—as well as to smaller coastal communities such as Fogo Island, Twillingate, and Bonavista, where radar, weather, and coastal defence installations formed part of the wider North Atlantic operational network, the expedition contributes to a distributed, place-based visitor experience grounded in historical interpretation, education, and community engagement.

7. Training and Academic Collaboration

The expedition will support the involvement of students and researchers through Memorial University’s Marine Institute.

This includes:

- Participation in field operations
- Exposure to advanced survey and imaging technologies
- Integration of expedition data into academic research projects

The project is intended not only as documentation, but as a training platform for the next generation of marine scientists and archaeologists.

8. Contribution to National Commemoration

The Bell Island attacks represent one of the few instances of direct enemy action within Canadian waters during the Second World War.

By documenting and interpreting this site, the expedition contributes to:

- Recognition of the Royal Canadian Navy, the Merchant Navy, Allied seafarers, and the Royal Canadian Air Force
- Greater public understanding of Canada’s and Newfoundland’s—then a separate Dominion—role in the Battle of the Atlantic
- The creation of a meaningful and enduring digital memorial

Unlike land battlefields, where physical markers often remain, the naval war is largely invisible. This project seeks to make that history visible again.

Looking Ahead

The Bell Island Battle of the Atlantic Expedition 2027 is not a single survey, but the foundation of a long-term effort to document, understand, and interpret Canada's wartime heritage in its full geographic context, from the seabed to the coastal and terrestrial landscape.

Through the integration of archaeology, marine science, and advanced imaging technologies, the project will transform a largely unseen battlefield into a fully accessible and enduring digital record, while supporting ongoing research, education, and public engagement.

Building on the approaches developed during NOAA's Battle of the Atlantic Expedition 2016 and the HMCS *Canada* Expedition 2025, the Bell Island Expedition 2027 extends this model to a larger and more complex environment, linking science, history, and commemoration within the broader North Atlantic context.

Technology and Methodology

What is Photogrammetry?

Photogrammetry is a technique that uses large numbers of overlapping photographs to extract precise measurements and generate detailed three-dimensional models of objects or environments. While traditionally used in aerial surveying, architecture, and engineering, advances in digital imaging and computing have made photogrammetry an essential tool in archaeology—particularly for documenting underwater sites.

In marine archaeology, photogrammetry enables the non-invasive recording of submerged cultural heritage, including shipwrecks and associated structures. By creating accurate digital replicas, researchers can analyze structural details, monitor changes over time, and share results widely without disturbing fragile underwater environments.

Underwater photogrammetry is conducted using waterproof cameras and high-intensity lighting systems. Divers or remotely operated vehicles capture hundreds or thousands of overlapping images from multiple angles. Specialized software then identifies common reference points and reconstructs the site in three-dimensional space, producing a high-resolution digital model.

These models allow archaeologists, marine scientists, and the public to explore underwater heritage sites virtually. They can be examined on-screen, rendered as physical models, or integrated into immersive virtual environments, significantly expanding access to sites that are otherwise difficult or impossible to visit.

Application to the Bell Island Expedition 2027

The Bell Island Expedition will apply photogrammetry at a significantly larger scale than previous work. All four wrecks will be documented using a combination of diver-based imaging

and remotely operated vehicle (ROV) systems, integrated with acoustic positioning and geospatial data to support the creation of a unified Digital Terrain Model (DTM).

In a comparative study, one wreck—*PLM-27*—will be surveyed using both diver-based photogrammetry and ROV-based imaging systems. This dual approach will allow direct comparison between human-operated and remotely operated survey methods in cold-water North Atlantic conditions, contributing to the refinement of best practices in underwater documentation.

VOYIS Imaging and Survey Technology

A central component of the expedition is the use of advanced underwater imaging systems developed by VOYIS, a Canadian firm specializing in subsea optical and survey technologies.

VOYIS systems integrate high-resolution imaging with precise spatial referencing, enabling efficient and repeatable documentation of complex underwater environments. These systems are designed for deployment on remotely operated platforms and are well suited to large-scale survey operations such as those planned for Bell Island.

VOYIS systems have been deployed in offshore energy, marine research, and defence applications, providing a proven platform for high-resolution subsea mapping in challenging environments. VOYIS imaging software—developed under the company’s earlier name, 2G Robotics—was used to support photogrammetric documentation of the German submarine *U-576* and the freighter *Bluefields* during NOAA’s Battle of the Atlantic Expedition 2016.

Memorial University’s Marine Institute

The expedition is conducted in partnership with the Memorial University of Newfoundland Marine Institute, which provides vessel support, remotely operated vehicle (ROV) systems, USBL positioning, and technical personnel. These capabilities enable integrated subsea survey operations and support the acquisition of high-resolution geospatial data required for the project’s Digital Terrain Model.

The Marine Institute will also provide a licensed drone operator to acquire aerial imagery of the Bell Island landscape, which will be integrated with subsea and terrestrial datasets to produce a unified Digital Terrain Model. This combination of vessel-based operations, aerial survey, and geospatial integration reflects the Institute’s role as a leading centre for applied ocean technology.

The Marine Institute also contributes an academic and educational dimension to the expedition, with students participating directly in field operations and survey workflows. This integration of research, training, and applied technology reflects the Institute’s mandate and provides hands-on experience in real-world marine environments.

Experience from the HMCS *Canada* Expedition

On the HMCS *Canada* Expedition 2025, photogrammetry served both archaeological and ecological purposes. The expedition produced a complete three-dimensional model of the wreck while also supporting ecological assessments, including habitat mapping and species monitoring.

This integrated approach—combining marine archaeology with marine biology and environmental observation—forms the methodological foundation for the Bell Island Expedition and will be extended to a larger and more complex battlefield environment.

Technology Developed for the Expedition: Photogrammetry Dive Sled

To address the challenges of deep-water photogrammetry at approximately 70 metres (235 feet), the HMCS *Canada* team developed a custom diver propulsion vehicle (DPV)-mounted imaging platform.

The system consists of a lightweight frame with integrated buoyancy control, high-output underwater lighting, and dual camera systems mounted to ensure consistent image overlap. Attached to a diver propulsion vehicle, the platform provides stability and control during imaging, allowing large areas to be surveyed efficiently under limited bottom time conditions.

Field testing demonstrated that the system maintained stability at depth, produced high-quality overlapping imagery, and enabled reliable three-dimensional reconstruction of complex structures. The platform also allowed imaging to be conducted at speeds significantly greater than conventional fin-based survey methods, improving overall efficiency.

This technology will be deployed and further refined during the Bell Island Expedition, where environmental conditions—depth, temperature, and currents—are comparable to those encountered during its development off Vancouver Island and its operational use on the HMCS *Canada* Expedition 2025.